LEARNING MODULE

3

Taylor Series and Approximation

How much is ≈ worth? Local polynomial models, their error, and the money they explain.

VOLUME I · Calculus for Finance **BUILDS ON** LM1 · LM2

LEADS TO LM4 · LM7 · LM14 · LM28 · LM42 · LM46 **CFA L1** Q1 Rates and Returns **TIME** ≈ 2.5 h + lab

LEARNING OUTCOMES

Mastery	*The candidate should be able to:*
☐	1 construct the Taylor polynomial of order n of a smooth function about a point, and interpret its terms as level, slope and curvature
☐	2 derive the expansions of e^x, $\ln(1+x)$ and $(1+x)^\alpha$, and state where each one converges
☐	3 bound the truncation error with the Lagrange remainder, and decide when a truncated expansion can be trusted
☐	4 relate simple and log returns, and arithmetic and geometric means, through second-order expansions, and quantify volatility drag
☐	5 decompose the long-run return of a daily-reset leveraged ETF into a power of the index, a variance drag and a residual, and estimate each term from data
☐	6 apply first- and second-order approximations to bond prices (duration and convexity), and recognise the same structure in option delta–gamma and in Itô's lemma
☐	7 implement and verify Taylor approximations in Python

LEARNING MODULE OVERVIEW

- A Taylor polynomial is the best local polynomial model of a smooth function: it matches the function's value and its first n derivatives at one point.
- Three expansions — e^x, $\ln(1+x)$ and $(1+x)^\alpha$ — generate most approximations used in finance. The series for $\ln(1+x)$ converges only for $-1 < x \le 1$.
- The Lagrange remainder turns "≈" into a statement with an error bound. The error of an order-n approximation shrinks like $|x|^{n+1}$.
- Log and simple returns differ by $R^2/2$ to second order. Over a sample, geometric return ≈ arithmetic mean − half the variance: this is volatility drag, and it is an approximation, not an identity.
- A daily-reset $L\times$ fund earns approximately the L-th power of the index's growth, taxed by $\exp\big(-\frac{L(L-1)}{2}\sum R_t^2\big)$. On twelve years of 0050 data, the second-order term matches the exact rebalancing effect to within 0.0001 log points.
- "Leveraged ETFs decay" and "leveraged ETFs beat $L\times$" are both true, against different benchmarks. In strong trends a 2× fund can more than double the index's return.
- Duration–convexity, delta–gamma and Itô's lemma are all second-order Taylor expansions. Itô's lemma is the one in which the second-order term does not vanish.

1 INTRODUCTION

Almost every formula a trader uses daily is an approximation in disguise: "log return ≈ simple return", "geometric mean ≈ arithmetic mean − $\sigma^2/2$", "a bond's price change ≈ −duration × yield change". Each is the first or second term of a **Taylor expansion**. Using them well requires knowing three things: where they come from, how wrong they are, and when they break. This module answers all three, and then uses the answers on real money.

MOTIVATING CASE **Two numbers that cannot both be true**

From the close of 31 October 2014 to 8 October 2026, Yuanta Taiwan 50 (0050) returned **+902%** including dividends: one unit grew to 10.02. Over the same days the daily-reset 2× fund on the same index, Yuanta Taiwan 50 Bull 2X (00631L), returned **+4,354%**: one unit grew to 44.54. "Twice the index's return" would have been +1,804%. The leveraged fund delivered about 2.4 times *twice* the index.

Then take calendar 2022 in the US. QQQ lost 32.6%; the 3× fund TQQQ lost 79.1%. Three times QQQ's loss is −97.7%, so TQQQ lost *less* than three times the index.

Yet every regulator and textbook warns that leveraged ETFs **decay**. How can a fund decay and beat its multiple at the same time? How large is the decay, and what sets its size? By Section 7 you will be able to split each fund's return into three numbers and reproduce the result to four decimals.

Exhibit 1: The leveraged-fund paradox in two markets

Period	Index fund	Leveraged fund	"Multiple × index return"
2014-10-31 → 2026-10-08 (11.9 years)	0050: +902.25%	00631L (2×): +4,354.5%	2 × 902.25% = +1,804.5%
Annualized	21.3%	37.4%	—
Calendar 2022	QQQ: −32.58%	TQQQ (3×): −79.09%	3 × (−32.58%) = −97.7%

Source: FinMind `TaiwanStockPrice`, `TaiwanStockDividendResult` and `USStockPrice`, retrieved 10 October 2026; 0050 includes reinvested cash dividends and is adjusted for its 4-for-1 split (18 June 2025); 00631L is adjusted for its 22-for-1 split (31 March 2026); QQQ and TQQQ use adjusted closes. Computations: companion notebook `LM03_lab.ipynb`.

The tool that resolves the paradox is a second-order Taylor expansion of $\ln(1+x)$. Sections 2–4 build the tool and its error bound. Sections 5–7 apply it to returns, means and leverage. Section 8 shows the same idea in bond pricing, option hedging and, as a preview of LM28, Itô's lemma.

> **NOTE** **Notation used throughout the book**
>
> R is a **simple return** for one period, $R = P_t/P_{t-1} - 1$ (plus income). $r = \ln(1+R)$ is the corresponding **log return**. μ and σ are the mean and standard deviation of R. g is a **log growth rate** per period and G a **growth factor** (end value ÷ start value). We write $=$ only for exact statements, $\approx$ for approximations whose neglected order is stated, and $:=$ for definitions. In earlier notes we wrote r for a simple return; from now on r always means a log return.

2 THE TAYLOR POLYNOMIAL: A LOCAL MODEL

- ☐ construct the Taylor polynomial of order n of a smooth function about a point, and interpret its terms as level, slope and curvature

Tangent line, then curvature

Near a point a, a smooth function looks like a straight line: its tangent. The tangent line is the first-order model

$$f(x) \approx f(a) + f'(a)\,(x-a).$$

For $f(x) = \ln(1+x)$ at $a = 0$, $f(0) = 0$ and $f'(0) = 1$, so $\ln(1+x) \approx x$. That is the familiar "log return ≈ simple return". It is exact at $x = 0$ and drifts away as x grows, because $\ln(1+x)$ bends downward. A better model adds the bend, measured by the second derivative:

$$f(x) \approx f(a) + f'(a)\,(x-a) + \tfrac{1}{2} f''(a)\,(x-a)^2.$$

With $f''(x) = -1/(1+x)^2$ and $f''(0) = -1$ this gives $\ln(1+x) \approx x - x^2/2$. The $-x^2/2$ term is the curvature correction, and it is the source of every "$-\sigma^2/2$" in this book.

DEFINITION **Taylor polynomial**

Let f have n derivatives at a. The **Taylor polynomial of order n of f about a** is

$$T_n(x) := \sum_{k=0}^{n} \frac{f^{(k)}(a)}{k!}\,(x-a)^k = f(a) + f'(a)(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \cdots + \frac{f^{(n)}(a)}{n!}(x-a)^n \tag{3.1}$$

When $a = 0$ it is also called a **Maclaurin polynomial**. Letting $n \to \infty$ gives the **Taylor series**, when that series converges.

Why the coefficients are $f^{(k)}(a)/k!$

T_n is built so that it agrees with f at a in value, slope, curvature and every higher derivative up to order n. Write a general polynomial $p(x) = \sum_{k=0}^{n} c_k (x-a)^k$ and differentiate it j times. Every term with $k < j$ disappears, every term with $k > j$ still contains a factor $(x-a)$ and vanishes at $x = a$, and the term $k = j$ becomes $j!\,c_j$. So

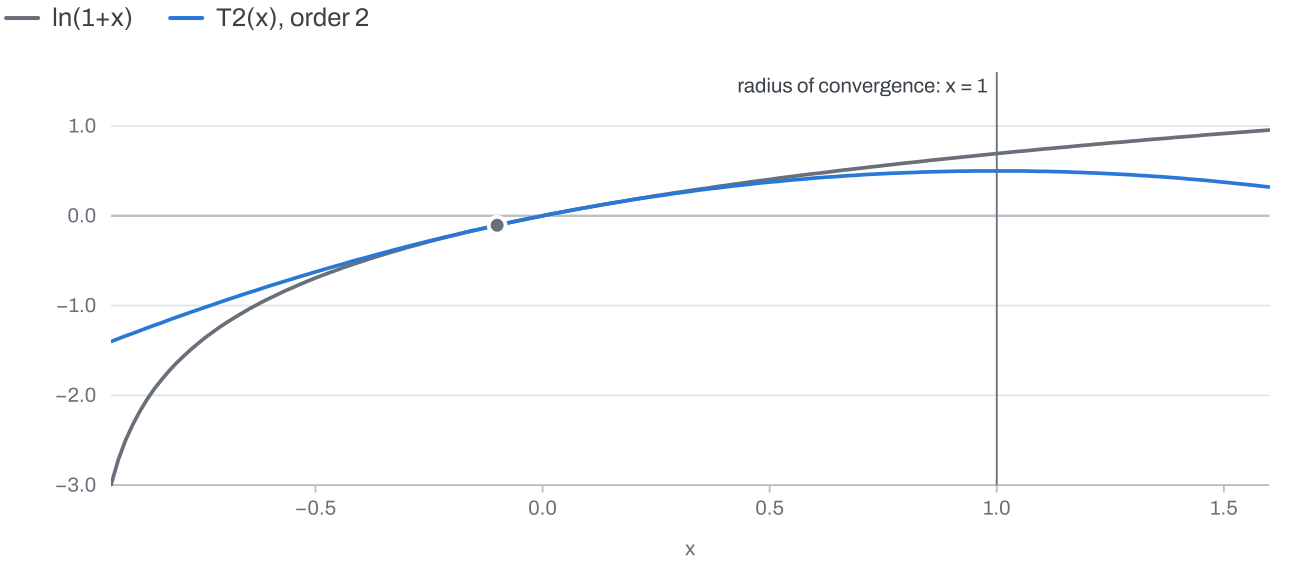
$$p^{(j)}(a) = j!\,c_j.$$

Requiring $p^{(j)}(a) = f^{(j)}(a)$ for $j = 0, \ldots, n$ forces $c_j = f^{(j)}(a)/j!$. The factorial is not decoration: it undoes the factorial that repeated differentiation of $(x-a)^j$ produces.

Each term has a job. The constant fixes the **level**, the linear term the **slope**, the quadratic term the **curvature**, and higher terms correct asymmetry and tails. In finance the terms through second order usually suffice, which is why "level, slope, curvature" recurs in duration–convexity, delta–gamma and the volatility drag.

Taylor polynomial explorer

INTERACTIVE



f(x)
−0.10536

T2(x)
−0.10500

Error f − T
−0.00036

Lagrange bound
0.00046

Grey curve: the function. Coloured curve: its order-n Taylor polynomial about 0. Raise n and watch the fit improve inside the radius of convergence and fail outside it.

EXAMPLE 1

Approximating $\ln(1.05)$ and $e^{0.05}$

A stock returns +5% in a month. Approximate its log return $\ln(1.05)$ and the growth factor $e^{0.05}$ of a 5% continuously compounded rate with Taylor polynomials of order 1, 2 and 3 about 0, and report each error.

SOLUTION

For $\ln(1+x)$ with $x = 0.05$: $T_1 = 0.05$, $T_2 = 0.05 - 0.00125 = 0.04875$, $T_3 = 0.04875 + 0.0000417 = 0.0487917$. The exact value is $\ln 1.05 = 0.0487902$, so the errors $f - T_n$ are -0.0012098, $+0.0000402$ and -0.0000015.

For e^x: $T_1 = 1.05$, $T_2 = 1.05125$, $T_3 = 1.0512708$ against the exact 1.0512711, with errors $+0.0012711$, $+0.0000211$ and $+0.0000003$.

The error after T_n is roughly the first omitted term, so each extra order shrinks it by the ratio of successive terms. For $\ln(1+x)$ that ratio is $x\,(n+1)/(n+2)$, here about 1/30 and 1/27. For e^x it is $x/(n+2)$, about 1/60 and 1/80, because the factorials make its terms shrink faster. That ratio is the practical rule for how many terms you need.

導讀 **泰勒多項式在做什麼**

把一個彎曲的函數，在某一點附近用多項式來近似：常數項對準高度，一次項對準斜率，二次項對準彎曲程度。在收斂範圍內，階數越高、離展開點越近，近似得越好。金融裡最重要的是二次項，因為 $\ln(1+x)$ 往下彎，所以「對數報酬 ≈ 簡單報酬」永遠有一個 $-x^2/2$ 的修正，後面所有的 $-\sigma^2/2$ 都從這裡來。

KNOWLEDGE CHECK 1

In $T_2(x) = f(a) + f'(a)(x-a) + \frac{1}{2}f''(a)(x-a)^2$, which coefficient would you change to make the approximation bend upward instead of downward, and what property of f does it encode?

ANSWER

The quadratic coefficient $\frac{1}{2}f''(a)$. Its sign is the sign of the curvature: $f''(a) > 0$ (convex) bends up, $f''(a) < 0$ (concave) bends down. $\ln(1+x)$ is concave, so its quadratic term is negative.

3 FOUR EXPANSIONS THAT RUN FINANCE

- [ ] derive the expansions of e^x, $\ln(1+x)$ and $(1+x)^\alpha$, and state where each one converges

Three Maclaurin series, plus the geometric series they all lean on, generate most approximations in this book.

The exponential. Every derivative of e^x is e^x, so $f^{(k)}(0) = 1$ for all k and

$$e^x = \sum_{k=0}^{\infty} \frac{x^k}{k!} = 1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \cdots \qquad \text{for all real } x. \tag{3.2}$$

The logarithm. Differentiate $f(x) = \ln(1+x)$ repeatedly: $f'(x) = (1+x)^{-1}$, $f''(x) = -(1+x)^{-2}$, $f'''(x) = 2(1+x)^{-3}$, and in general $f^{(k)}(x) = (-1)^{k+1}(k-1)!\,(1+x)^{-k}$. At 0 this gives $f^{(k)}(0)/k! = (-1)^{k+1}/k$, so

$$\ln(1+x) = \sum_{k=1}^{\infty} \frac{(-1)^{k+1}x^k}{k} = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \cdots \qquad \text{for } -1 < x \le 1. \tag{3.3}$$

A second derivation explains the range. For $|t| < 1$ the geometric series gives $\frac{1}{1+t} = 1 - t + t^2 - t^3 + \cdots$. Integrating from 0 to x term by term yields (3.3) for $|x| < 1$. At $x = 1$ the alternating series still converges, to $\ln 2$ (by Abel's theorem), which is why the range is $-1 < x \le 1$.

The binomial series. For any real exponent α, $f(x) = (1+x)^\alpha$ has $f^{(k)}(0) = \alpha(\alpha-1)\cdots(\alpha-k+1)$, so

$$(1+x)^\alpha = \sum_{k=0}^{\infty} \binom{\alpha}{k} x^k = 1 + \alpha x + \frac{\alpha(\alpha-1)}{2}x^2 + \cdots \qquad \text{for } |x| < 1, \tag{3.4}$$

where $\binom{\alpha}{k} = \alpha(\alpha-1)\cdots(\alpha-k+1)/k!$. With $\alpha = -1$ it is the alternating geometric series $1/(1+x) = 1 - x + x^2 - \cdots$. With $\alpha = \frac{1}{2}$ it gives $\sqrt{1+x} \approx 1 + \frac{x}{2} - \frac{x^2}{8}$. When α is a positive integer the series stops and (3.4) is the ordinary binomial theorem.

Exhibit 2: The expansions most used in finance

Function	Series about 0	Converges for	Typical use
e^x	$1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \cdots$	all x	continuous compounding, $R = e^r - 1$
$\ln(1+x)$	$x - \frac{x^2}{2} + \frac{x^3}{3} - \cdots$	$-1 < x \le 1$	log returns, growth rates, Kelly
$(1+x)^\alpha$	$1 + \alpha x + \frac{\alpha(\alpha-1)}{2}x^2 + \cdots$	$\lvert x\rvert < 1$	annualizing, real rates, leverage
$\frac{1}{1-x}$	$1 + x + x^2 + x^3 + \cdots$	$\lvert x\rvert < 1$	annuities and perpetuities (LM6)

The radius of convergence has a financial meaning

The series (3.3) diverges for $x > 1$ and the function itself does not exist at $x = -1$. Both edges matter. At $x = -1$, a return of −100%, the log of wealth is $-\infty$: that is **ruin**, and it is why log-utility investors such as Kelly bettors (LM41) never accept a position that can lose everything. For $x > 1$, $\ln(1+x)$ is perfectly finite, but no number of series terms will reach it. A truncated series is a local tool; for large moves, compute the exact function.

> **PITFALL** **Series approximations for large returns**
>
> For a +150% year, $x - x^2/2 = 1.5 - 1.125 = 0.375$, while $\ln 2.5 = 0.916$. Adding terms makes it worse, because the series diverges for $x > 1$. Use $\ln(1+R)$ itself whenever $|R|$ is not small. A rough working limit for the second-order formula is $|R| \le 0.1$, where its error stays below 4 bp of log return (Section 4).

EXAMPLE 2

Real returns and the Fisher approximation

A one-year deposit pays a nominal rate $i = 1.725\%$ while expected inflation is $\pi = 1.8\%$. Compute the real rate exactly and with first- and second-order approximations.

SOLUTION

Exactly, $1 + R_{\text{real}} = (1+i)/(1+\pi)$, so $R_{\text{real}} = 1.01725/1.018 - 1 = -0.07367\%$.

Expanding $1/(1+\pi) = 1 - \pi + \pi^2 - \cdots$ gives $R_{\text{real}} = (i-\pi)(1 - \pi + \pi^2 - \cdots)$. To first order $R_{\text{real}} \approx i - \pi = -0.07500\%$. Keeping the next term, $R_{\text{real}} \approx (i-\pi) - \pi(i-\pi) = -0.07365\%$, which matches the exact value to within 0.0001 percentage points. The first-order rule $i - \pi$ is the Fisher approximation of CFA Level I.

KNOWLEDGE CHECK 2

Using (3.4), write the second-order approximation of $(1+x)^{1/12}$ and use it to convert an annual return of 9% into a monthly geometric return.

ANSWER

$(1+x)^{1/12} \approx 1 + \frac{x}{12} + \frac{\frac{1}{12}(\frac{1}{12}-1)}{2}x^2 = 1 + \frac{x}{12} - \frac{11}{288}x^2$. With $x = 0.09$: $1 + 0.0075 - 0.000309 = 1.007191$, so about 0.7191% per month; the exact value $1.09^{1/12} - 1 = 0.7207\%$. The neglected third-order term is positive, which is why the approximation is slightly low.

4 HOW GOOD IS "≈"? THE REMAINDER TERM

- [ ] bound the truncation error with the Lagrange remainder, and decide when a truncated expansion can be trusted

An approximation without an error estimate is a guess. Taylor's theorem supplies the estimate.

THEOREM Taylor's theorem with Lagrange remainder

Let f have $n+1$ continuous derivatives on an interval containing a and x. Then

$$f(x) = T_n(x) + R_n(x), \qquad R_n(x) = \frac{f^{(n+1)}(\xi)}{(n+1)!}\,(x-a)^{n+1} \tag{3.5}$$

for some ξ strictly between a and x.

Read (3.5) as: *the error is the next term of the series, evaluated at an unknown point ξ*. We rarely know ξ, but we usually know how large $f^{(n+1)}$ can be on the interval, and that bounds the error. In **big-O notation** we write $R_n(x) = O\big(|x-a|^{n+1}\big)$: halve the distance to a and an order-n error shrinks by a factor of about 2^{n+1}.

The error of "log return ≈ $R - R^2/2$"

For $f(x) = \ln(1+x)$, $f'''(\xi) = 2/(1+\xi)^3$, so with $n = 2$ and $a = 0$

$$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3(1+\xi)^3}.$$

For $x > 0$, $\xi \in (0, x)$ gives $1 + \xi > 1$, and the error lies in $\left(0,\ x^3/3\right)$. For $-1 < x < 0$, $\xi \in (x, 0)$ gives $1 + \xi > 1 + x$, and the error is negative with size at most $|x|^3/\left(3(1+x)^3\right)$. Losses are approximated less accurately than gains of the same size, because $\ln(1+x)$ bends harder on the left.

Exhibit 3: Second-order approximation of ln(1+x): actual error and Lagrange bound

x	$\ln(1+x)$	$T_2 = x - x^2/2$	Error $f - T_2$	Lagrange bound
−0.20	−0.223144	−0.220000	−0.003144	0.005208
−0.10	−0.105361	−0.105000	−0.000361	0.000457
−0.05	−0.051293	−0.051250	−0.000043	0.000049
−0.01	−0.010050	−0.010050	−0.0000003	0.0000003
+0.01	0.009950	0.009950	+0.0000003	0.0000003
+0.05	0.048790	0.048750	+0.000040	0.000042
+0.10	0.095310	0.095000	+0.000310	0.000333
+0.20	0.182322	0.180000	+0.002322	0.002667
+0.50	0.405465	0.375000	+0.030465	0.041667

Note: the bound is $|x|^3/3$ for $x > 0$ and $|x|^3/(3(1+x)^3)$ for $x < 0$. Every actual error lies inside its bound, and the bound is tight for small $|x|$.

EXAMPLE 3

The two worst days in 0050's history

On 5 August 2024, 0050 fell 9.130%. On 7 April 2025, after the US tariff announcement, it fell the full daily limit of 10.000%. For each day, compute the log return exactly and with T_1 and T_2, and check the error against the Lagrange bound.

SOLUTION

5 August 2024, $R = -0.091299$: exact $\ln(1+R) = -0.095740$. $T_1 = -0.091299$ is off by 44.4 bp; $T_2 = -0.095467$ is off by 2.7 bp; the bound is $0.091299^3/(3 \times 0.908701^3) = 3.4$ bp.

7 April 2025, $R = -0.100000$: exact -0.105361. T_1 is off by 53.6 bp; $T_2 = -0.105000$ is off by 3.6 bp, inside the bound of 4.6 bp.

Even on record crash days, the second-order formula is good to a few basis points of log return. The first-order formula is not: treating a −10% day as a −10% log return understates the damage by more than half a percentage point.

EXAMPLE 4

Why the “rule of 72” works

Money growing at rate R per year doubles in $T^* = \ln 2 / \ln(1+R)$ years. Expand the denominator to explain where 72 comes from and why it works best near 8%.

SOLUTION

From (3.3), $\ln(1+R) = R\big(1 - \frac{R}{2} + \frac{R^2}{3} - \cdots\big)$. Using $1/(1-u) \approx 1 + u$ with $u = \frac{R}{2} - \frac{R^2}{3}$:

$$T^* \approx \frac{\ln 2}{R}\left(1 + \frac{R}{2}\right) = \frac{0.693}{R} + 0.347.$$

At $R = 6\%$ this gives 11.899 years against the exact 11.896. Writing the formula as $k/(100R)$ requires $k = 69.3 + 34.7R$, which equals 72 at $R \approx 7.8\%$. The rule of 72 is a second-order Taylor approximation tuned for returns near 8%.

DEEP DIVE **Proof of Taylor's theorem with the Lagrange remainder** optional · click to expand

Step 1 (integral form). By the fundamental theorem of calculus, $f(x) = f(a) + \int_a^x f'(t)\,dt$. Integrate by parts with $u = f'(t)$ and $dv = dt$, choosing the antiderivative $v = -(x-t)$:

$$\int_a^x f'(t)\,dt = \Big[-f'(t)(x-t)\Big]_a^x + \int_a^x f''(t)(x-t)\,dt = f'(a)(x-a) + \int_a^x f''(t)(x-t)\,dt.$$

Repeating n times gives $f(x) = T_n(x) + R_n(x)$ with

$$R_n(x) = \frac{1}{n!}\int_a^x f^{(n+1)}(t)\,(x-t)^n\,dt.$$

Step 2 (Lagrange form). Suppose $x > a$ (the other case is symmetric). The weight $(x-t)^n$ is non-negative on $[a, x]$ and $f^{(n+1)}$ is continuous, so by the mean value theorem for integrals there is $\xi \in (a, x)$ with

$$R_n(x) = \frac{f^{(n+1)}(\xi)}{n!}\int_a^x (x-t)^n\,dt = \frac{f^{(n+1)}(\xi)}{(n+1)!}(x-a)^{n+1}. \qquad \blacksquare$$

The integral form is exact and contains no unknown point, which makes it convenient when a sharper bound is needed.

導讀 **「≈」到底差多少**

泰勒定理說：誤差＝「下一項」，只是導數要在某個不知道的點 ξ 上取值。我們不知道 ξ 在哪，但能估出導數的上限，所以能給出誤差上界。以二階的 $\ln(1+x) \approx x - x^2/2$ 為例，連 2025/4/7 跌停（−10%）這種極端日，誤差也只有 3.6 bp；但若用一階的「對數報酬 ≈ 簡單報酬」，誤差會超過 50 bp。另外注意：同樣幅度，**下跌的誤差比上漲大**，因為 ln 在左邊彎得更厲害。

KNOWLEDGE CHECK 3

The fourth-order Taylor polynomial of e^x about 0 is used to compute $e^{-0.3}$. Bound the error.

ANSWER

$R_4 = e^{\xi}x^5/5!$ with $\xi \in (-0.3, 0)$, so $e^{\xi} < 1$ and $|R_4| < 0.3^5/120 = 2.03 \times 10^{-5}$. The actual error is $e^{-0.3} - T_4(-0.3) = 0.7408182 - 0.7408375 = -1.93 \times 10^{-5}$, inside the bound.

5 SIMPLE RETURNS AND LOG RETURNS

- [ ] relate simple and log returns, and arithmetic and geometric means, through second-order expansions, and quantify volatility drag

A holding-period (simple) return and a log return describe the same price move:

$$R_t = \frac{P_t + D_t}{P_{t-1}} - 1, \qquad r_t = \ln(1 + R_t).$$

From (3.3) and (3.2), each is a series in the other:

$$r = R - \frac{R^2}{2} + \frac{R^3}{3} - \cdots, \qquad R = e^r - 1 = r + \frac{r^2}{2} + \frac{r^3}{6} + \cdots. \tag{3.6}$$

Because $\ln(1 + x)$ is concave and touches the line $y = x$ at the origin, $\ln(1 + x) \le x$ for every $x > -1$, with equality only at 0. So the log return is always the smaller number, by about $R^2/2$.

Exhibit 4: Simple versus log returns

Simple return R	Log return $r = \ln(1 + R)$	Gap $R - r$	Second-order gap $R^2/2$
−50%	−69.31%	19.31%	12.50%
−20%	−22.31%	2.31%	2.00%
−10%	−10.54%	0.54%	0.50%
−1%	−1.005%	0.005%	0.005%
+1%	+0.995%	0.005%	0.005%
+10%	+9.53%	0.47%	0.50%
+20%	+18.23%	1.77%	2.00%
+50%	+40.55%	9.45%	12.50%
+100%	+69.31%	30.69%	50.00%

Note: the second-order gap is accurate below about 10% and useless beyond 50%; the asymmetry between ±50% is mostly the third-order term $R^3/3$.

Two aggregation rules, two return types

Each definition is additive along a different dimension, which is why both are needed.

- **Across time, log returns add.** $\ln(P_T/P_0) = \sum_{t=1}^{T} r_t$, because the log of a product is a sum. Simple returns compound instead: $1 + R_{0\to T} = \prod_t (1 + R_t)$.
- **Across assets, simple returns add.** A portfolio with weights w_i earns $R_p = \sum_i w_i R_i$ exactly. Log returns do not: $r_p = \ln\left(\sum_i w_i e^{r_i}\right) \neq \sum_i w_i r_i$.

The working rule follows. Use log returns for research along the time axis — trends, volatility, cross-asset comparisons. Use simple returns whenever money is aggregated across positions — portfolio accounting, settlement, drawdown in currency.

The gap in real data

Over 2,902 trading days from November 2014 to October 2026, 0050's mean daily simple return was 0.0871% and its mean daily log return 0.0794%. The difference, 0.0077% (0.77 bp) a day, is what (3.6) predicts: half the mean squared return is 0.771 bp, and half the variance is 0.767 bp. Summed over a year of about 243 trading days, this "small" daily gap is roughly 1.9 percentage points of annual return.

PYTHON

```python
import numpy as np
import polars as pl

df = pl.read_parquet("etf_daily_returns.parquet")          # columns: date, R_0050, R
R = df["R_0050"].to_numpy()
r = np.log1p(R)                                             # exact log returns
print(f"mean R   = {R.mean():.6%}")                         # 0.087120%
print(f"mean r   = {r.mean():.6%}")                         # 0.079422%
print(f"gap      = {(R.mean() - r.mean()) * 1e4:.3f} bp")   # 0.770 bp
print(f"E[R^2]/2 = {(R**2).mean() / 2 * 1e4:.3f} bp")       # 0.771 bp, eq. (3.6)
```

KNOWLEDGE CHECK 4

A portfolio holds 60% in asset A, which rose 10%, and 40% in asset B, which fell 10%. Compute the portfolio's simple return and log return, and the weighted average of the two log returns. Which aggregation is wrong?

ANSWER

$R_p = 0.6(0.10) + 0.4(-0.10) = 2.00\%$ exactly, so $r_p = \ln 1.02 = 1.980\%$. The weighted average of log returns is $0.6 \ln 1.1 + 0.4 \ln 0.9 = 0.6(0.0953) + 0.4(-0.1054) = 1.504\%$, which is wrong. Log returns do not aggregate across assets.

6 ARITHMETIC MEAN, GEOMETRIC MEAN AND VOLATILITY DRAG

- [ ] relate simple and log returns, and arithmetic and geometric means, through second-order expansions, and quantify volatility drag

Given returns $R_1, \ldots, R_T$, three averages answer three different questions.

$$\bar{R} = \frac{1}{T}\sum_{t=1}^{T} R_t, \qquad R_G = \Big(\prod_{t=1}^{T}(1+R_t)\Big)^{1/T} - 1, \qquad g = \ln(1+R_G) = \frac{1}{T}\sum_{t=1}^{T}\ln(1+R_t).$$

$\bar{R}$ is the average one-period outcome. R_G is the constant return that produces the same ending wealth. g is the same thing on the log scale, where it adds up over time. Wealth compounds, so R_G and g, not $\bar{R}$, describe what a buy-and-hold investor actually earned.

Deriving the drag

Apply (3.3) to each period and average:

$$g = \frac{1}{T}\sum_{t}\left(R_t - \frac{R_t^2}{2} + \frac{R_t^3}{3} - \cdots\right) = \bar{R} - \frac{1}{2}\cdot\frac{1}{T}\sum_{t} R_t^2 + \frac{1}{3}\cdot\frac{1}{T}\sum_{t} R_t^3 - \cdots$$

The mean of squares needs no probability theory. With the population variance $s^2 = \frac{1}{T}\sum_t (R_t - \bar{R})^2$, expanding the square gives the identity

$$\frac{1}{T}\sum_{t=1}^{T} R_t^2 = s^2 + \bar{R}^2. \tag{3.7}$$

Substituting and stopping at second order:

$$g \approx \bar{R} - \frac{s^2 + \bar{R}^2}{2}. \tag{3.8}$$

Converting back with (3.2), $R_G = e^g - 1 \approx g + \frac{g^2}{2}$, and $g^2 \approx \bar{R}^2$ to this order, so the $\bar{R}^2$ terms cancel:

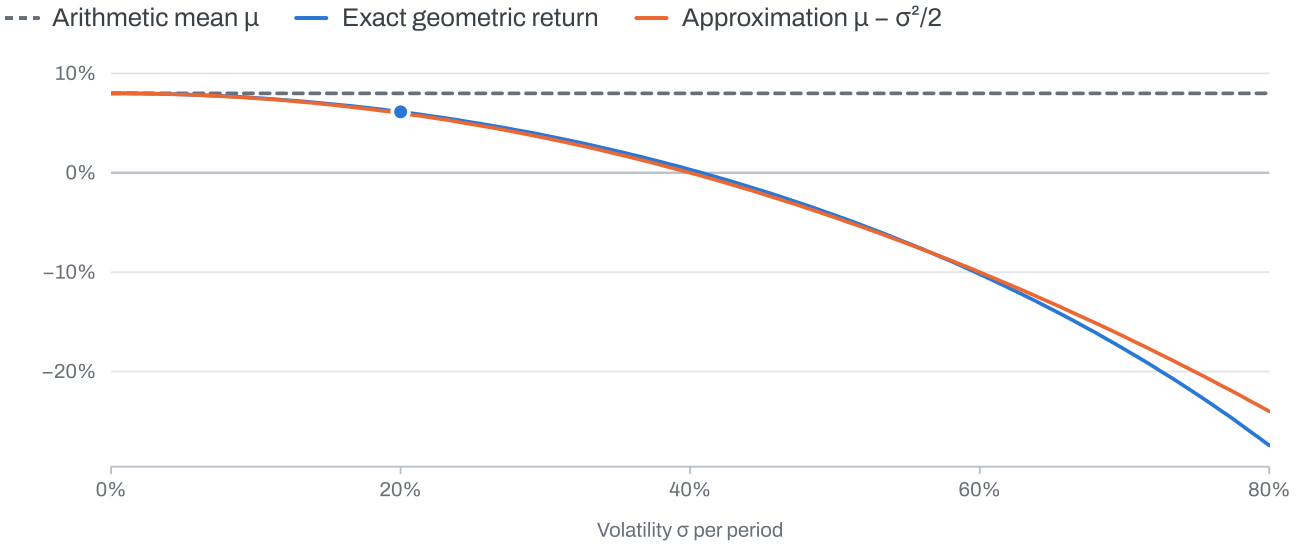
$$R_G \approx \bar{R} - \frac{s^2}{2}. \tag{3.9}$$

This is the CFA Level I statement "geometric mean ≈ arithmetic mean − σ²/2". The gap $\bar{R} - R_G \approx s^2/2$ is called **volatility drag**. Equations (3.8) and (3.9) are not the same formula: (3.9) approximates the geometric *simple* return, while (3.8) approximates the *log* growth rate and keeps an extra $-\bar{R}^2/2$. Both drop third- and higher-order terms, so both are approximations.

> **RESULT** **Volatility drag**
>
> To second order, the geometric mean return equals the arithmetic mean minus half the variance, $R_G \approx \bar{R} - s^2/2$, and the log growth rate is $g \approx \bar{R} - (s^2 + \bar{R}^2)/2$. The neglected terms are of third order in the returns. They grow with the length of the period, with the size of the moves, and with skewness.

Volatility drag in a two-outcome world

INTERACTIVE



Exact geometric	μ – σ²/2	Approx. error
6.132%	**6.000%**	**−13.2 bp**
Log growth g	**Mean wealth after N**	**Median wealth after N**
5.951%	**4.66×**	**3.29×**

Each period the return is μ + σ or μ − σ with equal probability, so σ is exactly the standard deviation. The exact geometric return is $\sqrt{(1+\mu)^2 - \sigma^2} - 1$.

How accuracy depends on the measurement period

The approximation error depends on how large the per-period returns are, so it depends on the sampling frequency. Exhibit 5 measures it on the same 0050 history at five frequencies.

Exhibit 5: Volatility-drag approximations on 0050, November 2014 – October 2026

Period	n	$\bar{R}$	s	R_G exact	$\bar{R} - \frac{s^2}{2}$	Error	g exact	$\bar{R} - \frac{s^2+\bar{R}^2}{2}$	Error
Daily	2,902	0.0871%	1.239%	0.0795%	0.0795%	−0.00 bp	0.0794%	0.0794%	−0.00 bp
Weekly	615	0.4065%	2.495%	0.3755%	0.3754%	−0.01 bp	0.3748%	0.3746%	−0.02 bp
Monthly	144	1.768%	5.649%	1.6135%	1.6087%	−0.5 bp	1.6006%	1.5931%	−0.7 bp
Quarterly	49	5.367%	10.953%	4.816%	4.767%	−4.9 bp	4.704%	4.623%	−8.0 bp
Annual	11	18.61%	20.17%	16.72%	16.57%	−14.9 bp	15.46%	14.84%	−62.1 bp

Note: total-return series with dividends reinvested; calendar periods, with partial first and last periods except for the annual row, which uses the full years 2015–2025. s is the population standard deviation, as in (3.7).

At daily frequency the approximation is exact to the displayed precision. At annual frequency, where single observations reach +48.7% and −21.3%, the error of (3.9) is 15 bp and that of (3.8) is 62 bp. The rule: second-order formulas are reliable when per-period standard deviations are a few percent, and need checking when they approach 20%.

EXAMPLE 5

Five years of 0050

0050's calendar-year total returns for 2020–2024 were 31.08%, 21.97%, −21.34%, 27.40% and 48.67%. Compute the arithmetic mean, the exact geometric mean and the approximation (3.9).

SOLUTION

$\bar{R} = (31.08 + 21.97 - 21.34 + 27.40 + 48.67)/5 = 21.557\%$. The population variance of the five returns is $s^2 = 0.05401$ ($s = 23.24\%$), so (3.9) gives $21.557\% - 2.701\% = 18.856\%$.

Exactly, $R_G = (1.3108 \times 1.2197 \times 0.7866 \times 1.2740 \times 1.4867)^{1/5} - 1 = 18.957\%$. The approximation is 10 bp low. An investor who quoted the 21.6% arithmetic mean as "what 0050 earned" would overstate the realized compound return by 2.6 percentage points a year.

EXAMPLE 6

The +2% / −1% bet, revisited

A strategy earns +2% or −1% per trade with equal probability and is traded with a fixed fraction of capital. Compute the arithmetic mean, the exact geometric return, the log growth rate, and both approximations.

SOLUTION

$\bar{R} = 0.5\%$ and $s^2 = \frac{1}{2}(0.015^2 + 0.015^2) = 0.000225$. Exactly, $R_G = \sqrt{1.02 \times 0.99} - 1 = 0.48881\%$ and $g = \frac{1}{2}(\ln 1.02 + \ln 0.99) = 0.48761\%$.

Equation (3.9) gives $0.5\% - 0.01125\% = 0.48875\%$, off by 0.006 bp. Equation (3.8) gives $0.5\% - (0.000225 + 0.000025)/2 = 0.48750\%$, off by 0.011 bp. With small moves both are excellent, but the drag is real: about 2.2% of the arithmetic edge disappears to compounding.

PITFALL **"Geometric = arithmetic − σ²/2" is not an identity**

Take returns of +100% and −50% with equal probability. The arithmetic mean is +25%, but $(2 \times 0.5)^{1/2} - 1 = 0\%$ exactly: wealth goes nowhere. The approximation gives $25\% - 0.75^2/2 = -3.1\%$. With moves this large the neglected third- and fourth-order terms dominate. For position sizing, compute $\frac{1}{T}\sum \ln(1 + R_t)$ directly rather than plugging into $\mu - \sigma^2/2$ — that is exactly the empirical-Kelly procedure of LM45.

導讀 **波動就是成本**

同樣的算術平均報酬，波動越大，複利後實際賺到的（幾何平均）越少，差額約等於 $s^2/2$，這就是「波動拖累」。它來自 $\ln(1+x)$ 的二次項，所以是二階近似：日報酬時幾乎完全精準；年報酬（單年 +48.7%、−21.3% 這種大波動）時就會有 15 到 62 bp 的誤差。要決定部位大小時，請直接算 $\frac{1}{T}\sum \ln(1 + R_t)$，不要只套 $\mu - \sigma^2/2$。另外記得：(3.9) 近似的是「幾何平均簡單報酬」R_G，(3.8) 近似的是「對數成長率」g，兩者差一個 $\bar{R}^2/2$。

KNOWLEDGE CHECK 5

Two funds have the same arithmetic mean annual return of 10%. Fund A has annual volatility 10%, fund B 30%. Estimate each fund's geometric mean return. Over 20 years, roughly how many times more wealth does A produce?

ANSWER

A: $10\% - 0.1^2/2 = 9.5\%$. B: $10\% - 0.3^2/2 = 5.5\%$. Over 20 years, $(1.095/1.055)^{20} = 2.11$: about twice the wealth, from identical arithmetic means.

7 LEVERAGED ETFS: POWER, DRAG AND RESIDUAL

- [ ] decompose the long-run return of a daily-reset leveraged ETF into a power of the index, a variance drag and a residual, and estimate each term from data

The daily-reset identity

A daily-reset fund with leverage L promises L times the index's *daily* simple return. Ignore costs for now and write the fund's daily return as LR_t. Its log return minus L times the index's log return is, by (3.3),

$$\ln(1+LR) - L\ln(1+R) = -\frac{L^2-L}{2}R^2 + \frac{L^3-L}{3}R^3 - \frac{L^4-L}{4}R^4 + \cdots \tag{3.10}$$

The first-order terms cancel exactly — that is what "$L\times$ daily" buys you. What remains starts at second order. Summing over the T days of a holding period and writing $G = S_T/S_0$ for the index's growth factor and $\mathrm{RV} = \sum_t R_t^2$ for its **realized variance**:

$$\frac{V_T}{V_0} \approx G^{\,L}\exp\!\Big(-\frac{L(L-1)}{2}\,\mathrm{RV}\Big). \tag{3.11}$$

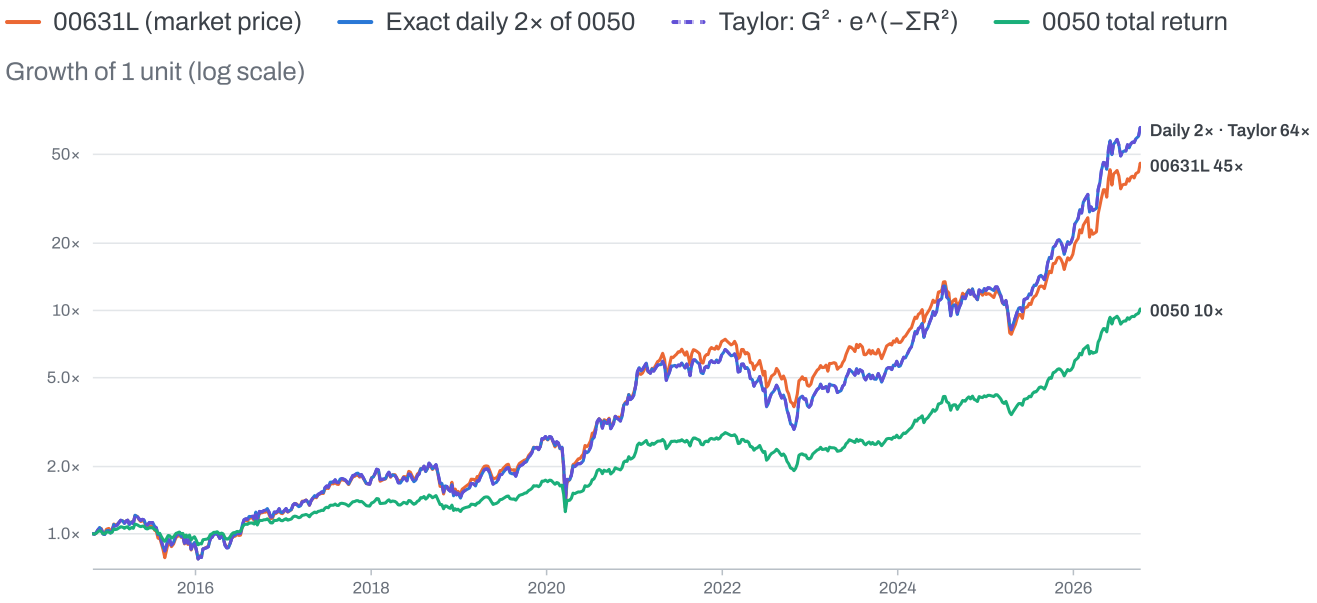
> **RESULT** **What a daily-reset fund delivers**
>
> Before costs, an $L\times$ daily fund returns approximately the L-th **power** of the index's growth factor, multiplied by a **variance drag** of $\exp(-\frac{L(L-1)}{2}\mathrm{RV})$. For $L=2$ the drag factor is $e^{-\mathrm{RV}}$; for $L=3$ it is $e^{-3\,\mathrm{RV}}$. The drag depends on the path only through realized variance (plus small third-order skewness terms).

This dissolves the paradox. "Twice the index's return" means $1+2(G-1)$. But for $L>1$ the power satisfies $G^L \geq 1+L(G-1)$, with equality only at $G=1$ (Bernoulli's inequality, itself a statement that x^L lies above its tangent line at $x=1$). So the fund has two forces: compounding of the trend, G^L versus $1+L(G-1)$, which favours the fund, and variance drag, which hurts it. Strong trends with low volatility favour the fund; choppy markets with high volatility punish it. Practice Problem 9 makes this precise: before costs, a 2× fund beats twice the index's simple return roughly when the trend $|1-1/G|$ exceeds the realized volatility $\sqrt{\mathrm{RV}}$.

Twelve years of 00631L

Exhibit 6 compares three quantities built from the same 2,902 days: 0050's total-return growth, the growth of an exact daily 2× of 0050 computed as $\prod_t(1+2R_t)$, and the Taylor formula $G^2e^{-\mathrm{RV}}$ evaluated along the path. 00631L's market price is plotted alongside.

Exhibit 6: Growth of one unit, November 2014 – October 2026 (log scale)



Note: the dashed Taylor curve lies on top of the exact daily-2× curve for the whole period; at the end they read 64.220 and 64.223. The gap between them and 00631L is not explained by (3.11): it is cost and replication basis.

Exhibit 7: Decomposing 00631L's log growth, 2014-10-31 → 2026-10-08

Component	Log points	Growth factor
$2 \times$ log growth of 0050 (2×2.30483)	4.6097	×100.45 ($= 10.0225^2$)
Variance drag, exact daily rebalancing $\sum[\ln(1+2R_t) - 2\ln(1+R_t)]$	−0.44729	×0.639
— second-order Taylor term $-\mathrm{RV}$	−0.44735	
— third-order Taylor $-\mathrm{RV} + 2\sum R_t^3$	−0.44461	
Residual: fees, financing, replication basis, price vs NAV	−0.3659	×0.694
00631L log growth	**3.7965**	**×44.54**

Note: the first four rows use only 0050 data. The residual is 00631L's actual log growth minus the exact daily-2× growth.

Three conclusions follow. First, the second-order Taylor term reproduces the exact rebalancing effect to within 0.0001 log points over twelve years: (3.11) is not a rough rule. Second, the drag averaged $0.4473/11.94 = 3.75\%$ a year, essentially 0050's realized annual variance ($0.1931^2 = 3.73\%$). Third, 00631L still grew 44-fold against 10-fold for 0050, because $G^2 = 100$ is a very large number: the trend term overwhelmed the drag.

The year-by-year view shows the two forces trading places.

Exhibit 8: Year-by-year decomposition of 00631L against twice 0050

Year	0050 total return	00631L	2 × 0050	Taylor drag $-\mathrm{RV}$	Exact rebalancing effect	Residual
2015	−6.31%	−16.54%	−12.62%	−0.0266	−0.0265	−0.0239
2016	19.64%	39.18%	39.29%	−0.0224	−0.0224	−0.0058
2017	18.13%	39.82%	36.27%	−0.0088	−0.0087	+0.0106
2018	−4.95%	−10.38%	−9.90%	−0.0270	−0.0280	+0.0200
2019	33.52%	70.87%	67.03%	−0.0152	−0.0150	−0.0273
2020	31.08%	68.07%	62.17%	−0.0528	−0.0527	+0.0306
2021	21.97%	62.22%	43.94%	−0.0311	−0.0308	+0.1174
2022	−21.34%	−36.44%	−42.68%	−0.0476	−0.0477	+0.0747
2023	27.40%	62.93%	54.80%	−0.0194	−0.0191	+0.0230
2024	48.67%	58.96%	97.34%	−0.0611	−0.0627	−0.2670
2025	36.85%	50.99%	73.71%	−0.0595	−0.0603	−0.1552

Note: drag, rebalancing effect and residual are in log points. The rebalancing effect differs most from $-\mathrm{RV}$ in 2018 and 2024, years with sharp sell-offs: the third-order term $2\sum R_t^3$ is negative when large down days dominate.

- **2017**, a steady uptrend with the lowest variance in the sample, is the textbook case for the power term: an exact daily 2× of 0050 would have returned 38.3% against twice the index's 36.3%.
- **2022** is the mirror image in a falling market: an exact daily 2× of 0050 would have lost 41.0%, less than twice 0050's 21.3% loss (42.7%). Compounding a downtrend also helps, because each day's 2× loss is taken on a smaller base. 00631L itself lost only 36.4%, helped by a positive residual.
- **2015**, a choppy, slightly negative year, is where drag wins: an exact daily 2× would have lost 14.5% and 00631L lost 16.5%, against twice the index's −12.6%.

What Taylor cannot see: the residual

The residual averaged −0.031 log points a year but swung from +0.117 (2021) to −0.267 (2024). It is not a constant fee. It contains the fund's management and trading costs, the financing embedded in its leverage, any gap between market price and NAV, and **replication basis**. 00631L builds most of its exposure with TAIEX futures rather than Taiwan 50 instruments. On 19 May 2026 it held TAIEX futures equal to 160.87% of net assets plus TSMC shares equal to 39.63%, while TSMC's weight was 43.7% in TAIEX and 60.9% in the Taiwan 50 index. When TSMC leads, as in 2024–2026, a futures-heavy fund lags twice the Taiwan 50; in years when the broader market does relatively better, the residual can turn positive. A model explains only the terms it contains: (3.11) prices the variance drag to the fourth decimal, and the residual needs its own model.

EXAMPLE 7

TQQQ in 2022

In 2022 QQQ's adjusted price fell 32.58% and TQQQ's fell 79.09%. QQQ's daily returns had realized variance $\mathrm{RV} = 0.1030$ (annualized volatility 32.1%). Decompose TQQQ's log return with (3.10) for $L = 3$.

SOLUTION

Index power: $3 \ln 0.67424 = 3 \times (-0.39417) = -1.18252$, so $G^3 = 0.3065$.

Variance drag: the second-order term is $-\frac{3 \cdot 2}{2}\mathrm{RV} = -3 \times 0.1030 = -0.3090$; adding the third-order term $8 \sum R_t^3$ gives -0.3107. The exact rebalancing effect $\sum[\ln(1 + 3R_t) - 3\ln(1 + R_t)]$ is -0.3132.

Residual: TQQQ's actual log return was $\ln(1 - 0.7909) = -1.56493$, so the residual is $-1.56493 - (-1.18252) - (-0.31325) = -0.0692$, about 6.7% of value. That is consistent with an expense ratio near 1% plus financing two extra units of exposure in a year when US policy rates rose from near zero to above 4%.

The paradox resolves cleanly: "3 × −32.6% = −97.7%" compares the fund to the wrong benchmark. The right benchmark is $G^3 e^{-3\,\mathrm{RV}} = 0.3065 \times 0.734 = 0.225$, a 77.5% loss before costs (77.6% with the exact rebalancing effect).

PITFALL **Two true statements about leveraged ETFs**

"Leveraged ETFs decay" is true *relative to* G^L: for a given index growth G, more realized variance means less money. In fact, by Bernoulli's inequality $1 + LR_t \le (1 + R_t)^L$ every day, so before costs a daily $L\times$ fund can never beat G^L. "Leveraged ETFs can beat $L\times$" is true *relative to* $1 + L(G - 1)$, because of the convexity of G^L. Arguments about leveraged funds usually go wrong because the two sides use different benchmarks. Neither statement says whether holding one is wise; that is a sizing question, answered by the Kelly analysis of LM42 and LM46.

The variance term in (3.11), $-\frac{L(L-1)}{2}\sigma^2$ per unit time, will reappear in LM42 as the curvature of the Kelly growth curve $g(f) = r_f + f(\mu - r_f) - \frac{f^2\sigma^2}{2}$. Leverage L and Kelly fraction f are the same variable seen from two sides.

導讀 **正二到底「耗損」多少**

每日重設的 L 倍基金，長期成長倍數 ≈ 指數成長倍數的 L **次方**，再乘上變異數拖累 $e^{-\frac{L(L-1)}{2}\mathrm{RV}}$。所以「正二會耗損」和「正二打敗兩倍報酬」都對，只是比較的基準不同：前者跟 G^2 比，後者跟 $1+2(G-1)$ 比。以 0050 這 12 年的日報酬計算：二階泰勒項 −0.44735 與精確的每日再平衡效果 −0.44729 幾乎一樣，平均每年拖累約 3.75%（≈ 0050 的年化變異數）。00631L 剩下每年約 −3.1% 的殘差不是泰勒能解釋的，來自費用、槓桿融資、市價與淨值的差距，以及「用台指期複製台灣50」造成的基差：台積電在台灣50權重 60.9%、在加權指數只有 43.7%（2026/5/19），台積電領漲的年份（2024–2026）正二就會落後兩倍台灣50。

KNOWLEDGE CHECK 6

An index ends a year exactly where it started ($G=1$) after a year with realized variance $\mathrm{RV}=0.04$. Before costs, what does a 2× daily fund return? A 3× fund? What does "L times the index's return" predict?

ANSWER

2×: $e^{-0.04}-1=-3.9\%$. 3×: $e^{-3\times 0.04}-1=-11.3\%$. "$L$ times 0%" predicts 0% for both. In a flat, volatile market, drag is all that remains.

8 THE SAME SECOND-ORDER IDEA ELSEWHERE

- [ ] apply first- and second-order approximations to bond prices (duration and convexity), and recognise the same structure in option delta–gamma and in Itô's lemma

Bond prices: duration and convexity

A bond's price $P(y)$ is a smooth, convex function of its yield. A second-order expansion about the current yield gives the standard approximation

$$\frac{\Delta P}{P}\approx -D_{\mathrm{mod}}\,\Delta y+\frac{1}{2}C\,(\Delta y)^2,\qquad D_{\mathrm{mod}}:=-\frac{1}{P}\frac{dP}{dy},\quad C:=\frac{1}{P}\frac{d^2P}{dy^2}. \tag{3.12}$$

Modified duration is the slope term and convexity the curvature term: the same level–slope–curvature structure as Section 2.

EXAMPLE 8

A 10-year bond under a 100 bp shock

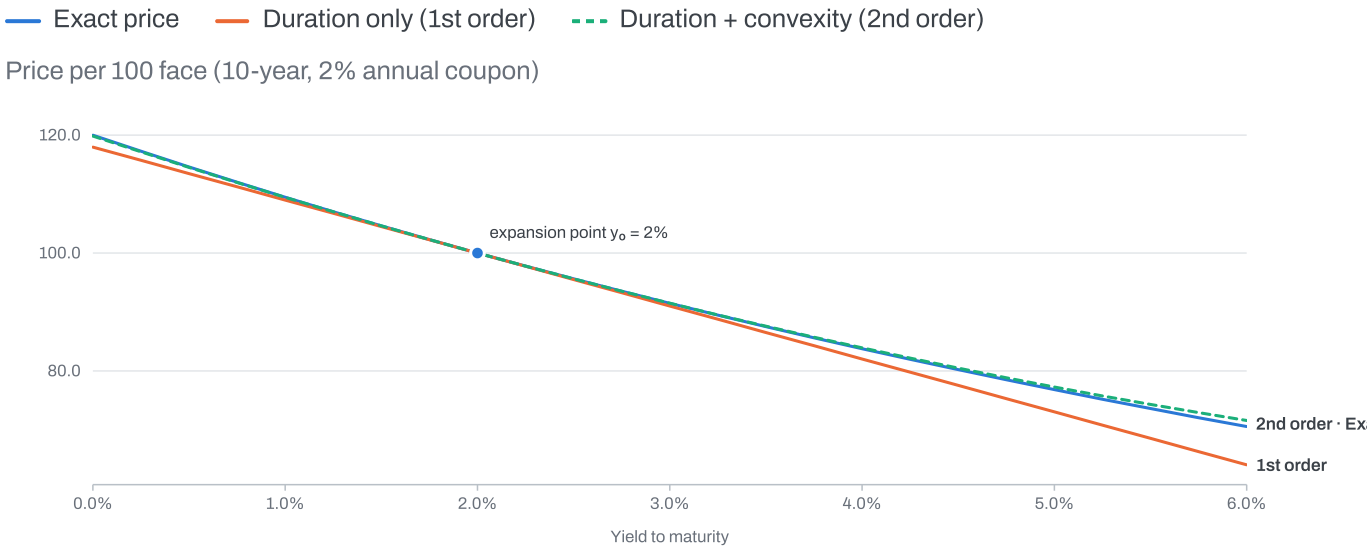
A 10-year bond pays a 2% annual coupon and trades at par (yield 2%). Its modified duration is 8.983 and its convexity 94.00. Estimate the percentage price change for yield changes of +100 bp, −100 bp and +300 bp using duration alone and duration plus convexity, and compare with exact repricing.

SOLUTION

Δy	Exact	Duration only	Duration + convexity
+100 bp	−8.530%	−8.983%	−8.513%
−100 bp	+9.471%	+8.983%	+9.453%
+300 bp	−23.165%	−26.948%	−22.718%

Duration alone is symmetric and misses the convexity gain on both sides. The second-order term cuts the error at ±100 bp from 0.45–0.49 points to about 0.02. At +300 bp the second-order error grows to 0.45 points: three times the shock gives roughly 25 times the error, close to the factor of 27 that an $O(\Delta y^3)$ remainder implies.

Exhibit 9: Price–yield curve with its first- and second-order approximations



Note: 10-year bond, 2% annual coupon, expanded about $y_0 = 2\%$ with $D_{\text{mod}} = 8.983$ and $C = 94.00$.

Options: delta and gamma

An option's value $V(S, t)$ expanded to second order in the underlying price and first order in time gives the trader's daily P&L approximation

$$\Delta V \approx \Delta \cdot \Delta S + \tfrac{1}{2}\,\Gamma\,(\Delta S)^2 + \Theta\,\Delta t,$$

where $\Delta = \partial V/\partial S$, $\Gamma = \partial^2 V/\partial S^2$ and $\Theta = \partial V/\partial t$. A delta-hedged book removes the first term and keeps the gamma term: it earns $\tfrac{1}{2}\Gamma(\Delta S)^2$ from realized moves and pays Θ for the privilege. LM31 develops this for TXO and warrants.

A preview of Itô's lemma

Apply a second-order expansion to $\ln S$ when the price follows a random walk with per-period return $\mu\Delta t + \sigma\sqrt{\Delta t}\,Z$, with Z a standard random shock:

$$\Delta \ln S \approx \frac{\Delta S}{S} - \frac{1}{2}\Big(\frac{\Delta S}{S}\Big)^2.$$

In ordinary calculus the squared term is negligible, because $(\Delta x)^2$ is much smaller than Δx. Here it is not: $(\Delta S/S)^2 \approx \sigma^2 Z^2 \Delta t$ is of the same order as the drift $\mu\Delta t$, and averages to $\sigma^2\Delta t$. Keeping it yields

$$d\ln S = \Big(\mu - \frac{\sigma^2}{2}\Big)dt + \sigma\,dW. \tag{3.13}$$

This is Itô's lemma for the logarithm, and the $-\sigma^2/2$ in it is the volatility drag of Section 6. In discrete time the drag was an approximation; in continuous time it becomes exact, because every higher-order term vanishes as $\Delta t \to 0$. LM28 makes this rigorous, and LM29 uses it to solve geometric Brownian motion.

DEEP DIVE **Why $(\Delta W)^2$ behaves like Δt** optional · click to expand

Let $\Delta W = \sqrt{\Delta t}\,Z$ with Z standard normal. Then $\mathbb{E}[(\Delta W)^2] = \Delta t$ and $\mathrm{Var}[(\Delta W)^2] = \Delta t^2\,\mathrm{Var}(Z^2) = 2\Delta t^2$. Split $[0, T]$ into $n = T/\Delta t$ steps and add the squared increments: the sum has mean T and variance $n \cdot 2\Delta t^2 = 2T\Delta t$, which goes to 0 as $\Delta t \to 0$. The sum of squared increments therefore converges to the deterministic number T: the **quadratic variation** of Brownian motion. A smooth path has quadratic variation 0, because its squared increments are of order Δt^2. That single difference is why stochastic calculus keeps the second-order Taylor term and ordinary calculus discards it.

KNOWLEDGE CHECK 7

Why does a long-only bond portfolio with positive convexity gain from a large yield move in *either* direction, relative to the duration-only estimate?

ANSWER

The convexity term $\tfrac{1}{2}C(\Delta y)^2$ is positive whatever the sign of Δy. Rates up: the loss is smaller than duration predicts. Rates down: the gain is larger. It is the bond-market analogue of a long-gamma option position.

9 TAYLOR APPROXIMATIONS IN PYTHON

- [ ] implement and verify Taylor approximations in Python

Two habits make approximations safe in code. First, compute the exact quantity alongside the approximation and report the gap. Second, prefer numerically stable library functions: `np.log1p(x)` and `np.expm1(x)` are accurate for tiny `x`, where `np.log(1 + x)` loses digits to rounding.

PYTHON

```python
import math
import numpy as np

def taylor_log1p(x, n):
    """Order-n Maclaurin polynomial of ln(1+x), eq. (3.3)."""
    return sum((-1) ** (k + 1) * x**k / k for k in range(1, n + 1))

def lagrange_bound_log1p(x, n):
    """Upper bound on |ln(1+x) - T_n(x)| from eq. (3.5)."""
    base = 1.0 if x >= 0 else 1.0 + x
    return abs(x) ** (n + 1) / ((n + 1) * base ** (n + 1))

for x in (-0.10, -0.0913, 0.05, 0.50):
    err = math.log1p(x) - taylor_log1p(x, 2)
    print(f"x={x:+.4f}  error={err:+.2e}  bound={lagrange_bound_log1p(x, 2):.2e}")
```

The decomposition of Section 7 takes a few vectorized lines for any index–fund pair with aligned daily simple returns:

PYTHON

```python
import numpy as np
import polars as pl

def leverage_decomposition(df: pl.DataFrame, idx: str, fund: str, L: float) -> dict:
    """Split a daily-reset fund's log growth into power, drag and residual (eqs. 3.10
    R, RF = df[idx].to_numpy(), df[fund].to_numpy()
    power = L * np.log1p(R).sum()                                  # ln G^L
    exact_drag = (np.log1p(L * R) - L * np.log1p(R)).sum()         # exact rebalancing e
    taylor2 = -L * (L - 1) / 2 * (R**2).sum()                      # second-order term
    taylor3 = taylor2 + (L**3 - L) / 3 * (R**3).sum()              # plus third-order te
    actual = np.log1p(RF).sum()
    return {"power": power, "exact_drag": exact_drag, "taylor2": taylor2,
            "taylor3": taylor3, "residual": actual - power - exact_drag, "actual": ac

df = pl.read_parquet("etf_daily_returns.parquet")
print(leverage_decomposition(df, "R_0050", "R_00631L", L=2))
# power 4.6097, exact_drag -0.44729, taylor2 -0.44735, taylor3 -0.44461, residual -0.
```

IN PYTHON **Companion lab**

The notebook `LM03_lab.ipynb` rebuilds every exhibit in this module from the verified FinMind price files: corporate-action handling for the 0050 split and dividends and the 00631L split, the frequency study of Exhibit 5, the decompositions of Exhibit 6 to Exhibit 8, the bond example, and solutions to the computational practice problems. It is Polars-first and runs offline from the bundled data snapshot.

SUMMARY

- A Taylor polynomial (3.1) matches a function's value and first n derivatives at one point. Its coefficients are $f^{(k)}(a)/k!$ because differentiating $(x-a)^k$ k times produces $k!$.
- The expansions of e^x, $\ln(1+x)$ and $(1+x)^\alpha$ generate most financial approximations. The series for $\ln(1+x)$ converges only on $(-1, 1]$; for large moves, use the exact function.
- The Lagrange remainder (3.5) makes every "≈" checkable: the error is the next term at an unknown point. For $\ln(1+x) \approx x - x^2/2$ the error stays below 4 bp of log return for daily moves up to 10%, including the 2024 and 2025 crash days.
- Log returns add across time; simple returns add across assets. The log return is smaller than the simple return by about $R^2/2$.
- Volatility drag: $R_G \approx \bar{R} - s^2/2$ and $g \approx \bar{R} - (s^2 + \bar{R}^2)/2$. Both are second-order approximations whose error grows with the size of per-period moves.
- A daily-reset $L\times$ fund earns approximately $G^L e^{-L(L-1)\mathrm{RV}/2}$ before costs. On 0050 over 2014–2026 the Taylor term (−0.44735) matched the exact rebalancing effect (−0.44729). 00631L's further residual of

−3.1% a year reflects costs and futures-based replication.

- Duration–convexity, delta–gamma and Itô's lemma share the level–slope–curvature structure. In Itô's lemma the squared term survives because $(\Delta W)^2$ is of order Δt.

PRACTICE PROBLEMS

1. The second-order Maclaurin polynomial of $f(x) = e^{2x}$ is:

   A. $1 + 2x + x^2$

   B. $1 + 2x + 2x^2$

   C. $1 + x + \frac{x^2}{2}$

2. Using $\ln(1 + x) \approx x - x^2/2$, the log return corresponding to a simple return of +8% is *closest* to:

   A. 7.68%

   B. 7.70%

   C. 8.00%

3. Approximate $\ln(0.92)$ with the second-order Taylor polynomial about 0. Compute the Lagrange bound on the error and the actual error.

4. A fund's annual returns over three years are +30%, −20% and +10%. Its geometric mean return is *closest* to:

   A. 3.4%

   B. 4.6%

   C. 6.7%

5. Use the Lagrange form of the first-order remainder to prove that $\ln(1 + x) \leq x$ for every $x > -1$, with equality only at $x = 0$.

6. A strategy returns +60% or −40% per period with equal probability. Compute the arithmetic mean, the exact geometric mean and the approximation $\bar{R} - s^2/2$. Explain the size of the approximation error.

7. An index rises 10% on day 1 and falls 10% on day 2. Compute the two-day return of the index and of a 2× daily fund (before costs). Then compute G^2 and the approximation (3.11), and comment on the result.

8. Over a year an index's growth factor is $G = 1.20$ and its realized variance is $\mathrm{RV} = 0.09$. Before costs, a 3× daily fund's annual return is *closest* to:

   A. +32%

B. +60%

C. +73%

9. Show that, before costs and to first order in RV, a 2× daily fund outperforms twice the index's simple return over a period if and only if $|1 - 1/G| > \sqrt{\mathrm{RV}}$. Check the condition for 0050 in 2017 ($G = 1.1813$, $\mathrm{RV} = 0.0088$) and 2015 ($G = 0.9369$, $\mathrm{RV} = 0.0266$).

10. A bond has modified duration 7.5 and convexity 70. Estimate its percentage price change if its yield rises by 150 bp.

11. Which statement about the Maclaurin series of $\ln(1+x)$ is correct?

A. It converges for every real x.

B. It converges for $-1 < x \le 1$ only.

C. It converges for $|x| < 2$.

12. Derive the doubling-time approximation $T^* \approx 0.693/R + 0.347$ and use it for $R = 6\%$. Compare with the exact value and the rule of 72.

13. Nominal return is 3% and inflation is 2.5%. Compute the real return exactly, with the first-order Fisher approximation, and with the second-order correction.

14. *(Python)* Using `leverage_decomposition` from Section 9 and the lab data, compute the decomposition of 00631L for 2020–2023 combined. Which term changes sign compared with the full-period result?

SOLUTIONS

1. **B** is correct. $f(0) = 1$, $f'(0) = 2$ and $f''(0) = 4$, so $T_2 = 1 + 2x + \tfrac{4}{2}x^2 = 1 + 2x + 2x^2$. Equivalently, substitute $2x$ into (3.2).

2. **A** is correct. $0.08 - 0.08^2/2 = 0.08 - 0.0032 = 0.0768$. The exact value $\ln 1.08 = 0.07696$ differs by the third-order term.

3. $T_2(-0.08) = -0.08 - 0.0032 = -0.0832$; exactly, $\ln 0.92 = -0.0833816$. The error is -0.0001816. For $x < 0$ the bound is $|x|^3/\big(3(1+x)^3\big) = 0.000512/(3 \times 0.778688) = 0.000219$, so the actual error is about 83% of the bound.

4. **B** is correct. $(1.30 \times 0.80 \times 1.10)^{1/3} - 1 = 1.144^{1/3} - 1 = 4.586\%$. C is the arithmetic mean. As a check, $\bar{R} = 6.667\%$ and $s^2 = 0.04222$, so (3.9) gives $6.667\% - 2.111\% = 4.556\%$.

5. With $n = 1$ and $a = 0$, (3.5) gives $\ln(1+x) = x + \frac{f''(\xi)}{2}x^2 = x - \frac{x^2}{2(1+\xi)^2}$ for some ξ between 0 and x. Because $1 + \xi > 0$, the remainder is ≤ 0, and it equals 0 only if $x = 0$. Hence $\ln(1+x) \le x$, with

equality only at $x = 0$.

6. $\bar{R} = 10\%$, $s = 50\%$, so $\bar{R} - s^2/2 = 10\% - 12.5\% = -2.5\%$. Exactly, $\sqrt{1.6 \times 0.6} - 1 = \sqrt{0.96} - 1 = -2.020\%$. The error of 0.48 percentage points is large because the two outcomes, +60% and −40%, sit ±50% around the mean, where third- and fourth-order terms are not negligible (compare Exhibit 3 and Exhibit 4).

7. Index: $1.10 \times 0.90 - 1 = -1.00\%$. Fund: $1.20 \times 0.80 - 1 = -4.00\%$. $G^2 = 0.99^2 = 0.9801$, and $\mathrm{RV} = 0.1^2 + 0.1^2 = 0.02$, so (3.11) gives $0.9801\, e^{-0.02} = 0.9607$, a −3.93% return against the exact −4.00%. The fund lost four times as much as the index. Of its −0.0408 log return, −0.0201 is the power term $2 \ln 0.99$ and −0.0207 is drag, which the second-order term $-\mathrm{RV} = -0.0200$ captures almost entirely even with 10% daily moves.

8. **A** is correct. $G^3 e^{-3\,\mathrm{RV}} = 1.728 \times e^{-0.27} = 1.728 \times 0.7634 = 1.319$, about +32%. B is three times the index's simple return; C is $G^3 - 1$ without drag.

9. Before costs the fund's growth factor is $G^2 e^{-\mathrm{RV}} \approx G^2(1 - \mathrm{RV})$; twice the index's simple return gives $1 + 2(G - 1) = 2G - 1$. The fund wins when $G^2 - G^2\mathrm{RV} > 2G - 1$, i.e. $(G-1)^2 > G^2\mathrm{RV}$, i.e. $|1 - 1/G| > \sqrt{\mathrm{RV}}$. For 2017: $|1 - 1/1.1813| = 0.153 > \sqrt{0.0088} = 0.094$, so the fund should win; indeed exact daily 2× returned 38.3% against 36.3%. For 2015: $|1 - 1/0.9369| = 0.067 < \sqrt{0.0266} = 0.163$, so the fund should lose; exact daily 2× returned −14.5% against −12.6%.

10. $-7.5 \times 0.015 + \tfrac{1}{2} \times 70 \times 0.015^2 = -0.1125 + 0.0079 = -10.46\%$.

11. **B** is correct (Section 3). At $x = 1$ the series converges to $\ln 2$; at $x = -1$ the function is undefined and the series diverges.

12. See Example 4. At $R = 6\%$: $0.693/0.06 + 0.347 = 11.899$ years; exact $\ln 2 / \ln 1.06 = 11.896$; the rule of 72 gives 12.

13. Exact: $1.03/1.025 - 1 = 0.4878\%$. First order: $3\% - 2.5\% = 0.5\%$. Second order: $(i - \pi) - \pi(i - \pi) = 0.5\% - 0.0125\% = 0.4875\%$.

14. Over 2020–2023 the residual is $+0.2456$ log points (the sum of the four yearly residuals in Exhibit 8), against −0.3659 for the full period. The residual changes sign; the drag stays negative, as it must. The residual is driven largely by replication basis, which can help as well as hurt.

GLOSSARY

Term	中文	Meaning
Taylor polynomial	泰勒多項式	Polynomial matching a function's value and first n derivatives at a point, (3.1).
Maclaurin series	馬克勞林級數	Taylor series about 0.
Lagrange remainder	拉格朗日餘項	Error term $f^{(n+1)}(\xi)(x-a)^{n+1}/(n+1)!$ of an order-n approximation, (3.5).
Radius of convergence	收斂半徑	Distance from the expansion point within which a power series converges.
Big-O notation	大O符號	$O(h^k)$: a quantity bounded by a constant times h^k as $h \to 0$.
Simple return	簡單報酬	$R = (P_t + D_t)/P_{t-1} - 1$; adds across assets.
Log return	對數報酬	$r = \ln(1+R)$, the continuously compounded return; adds across time.
Geometric mean return	幾何平均報酬	Constant per-period return producing the same terminal wealth.
Volatility drag	波動拖累	Arithmetic minus geometric mean, $\approx s^2/2$ to second order.
Realized variance	實現變異數	$\mathrm{RV} = \sum_t R_t^2$ over a holding period.
Daily-reset leveraged ETF	每日重設槓桿型 ETF	Fund targeting L times the index's daily return.
Replication basis	複製基差	Return gap from tracking an index with different instruments, such as TAIEX futures for Taiwan 50.
Modified duration	修正存續期間	$-P'(y)/P$: first-order price sensitivity to yield.
Convexity	凸性	$P''(y)/P$: second-order price sensitivity to yield.
Delta, gamma	Delta、Gamma	First and second derivatives of an option's value with respect to the underlying price.
Quadratic variation	二次變分	Limit of the sum of squared increments of a path; T for Brownian motion on $[0,T]$.

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